

board.c

```
#include <stdio.h>
#include <string.h>
#include "types.h"
#include "functions.h"

void setProperty(GameState game[],
                 int index,
                 char name[],
                 PropertyGroup group,
                 int purchasePrice,
                 int baseRent,
                 int mortgageValue,
                 int houseCost,
                 int hotelCost)
{
    game[0].board[index].index = index;
    game[0].board[index].type = PROPERTY;

    strcpy(game[0].board[index].name, name);
    strcpy(game[0].board[index].property.name, name);

    game[0].board[index].property.group = group;
    game[0].board[index].property.purchasePrice = purchasePrice;
    game[0].board[index].property.baseRent = baseRent;
    game[0].board[index].property.mortgageValue = mortgageValue;
    game[0].board[index].property.houseCost = houseCost;
    game[0].board[index].property.hotelCost = hotelCost;

    /* nobody owns it yet */
    game[0].board[index].property.owner = -1;

    game[0].board[index].property.houses = 0;
    game[0].board[index].property.hotel = 0;
    game[0].board[index].property.mortgaged = 0;
    game[0].board[index].property.loanLocked = 0;

    game[0].board[index].property.insurance = NO_INSURANCE;
    game[0].board[index].property.insuranceRoundsLeft = 0;

    game[0].board[index].property.damaged = 0;
    game[0].board[index].property.repairCostOwed = 0;
    game[0].board[index].property.lostIncomeRoundsLeft = 0;

    game[0].board[index].property.age = 0;
    game[0].board[index].property.depreciation = 0;
    game[0].board[index].property.condition = 100;

    game[0].board[index].property.roundsSinceMaintenance = 0;
    game[0].board[index].property.structurallyDamaged = 0;
    game[0].board[index].property.preDamagePurchasePrice = 0;
    game[0].board[index].property.preDamageBaseRent = 0;
    game[0].board[index].property.maintenanceCostMultiplierPercent = 100;
}

void setRailway(GameState game[], int index, char name[], int purchasePrice)
{
    game[0].board[index].index = index;
    game[0].board[index].type = RAILWAY;

    strcpy(game[0].board[index].name, name);
```

```

strcpy(game[0].board[index].property.name, name);

game[0].board[index].property.purchasePrice = purchasePrice;
game[0].board[index].property.mortgageValue = purchasePrice / 2;
game[0].board[index].property.owner = -1;
game[0].board[index].property.mortgaged = 0;
game[0].board[index].property.loanLocked = 0;

/* railways can't have houses/hotels/insurance, but we still zero
   these out since GameState is local now and won't do it for us */
game[0].board[index].property.houses = 0;
game[0].board[index].property.hotel = 0;
game[0].board[index].property.insurance = NO_INSURANCE;
game[0].board[index].property.insuranceRoundsLeft = 0;
game[0].board[index].property.damaged = 0;
game[0].board[index].property.repairCostOwed = 0;
game[0].board[index].property.lostIncomeRoundsLeft = 0;
game[0].board[index].property.age = 0;
game[0].board[index].property.depreciation = 0;
game[0].board[index].property.condition = 100;
game[0].board[index].property.roundsSinceMaintenance = 0;
game[0].board[index].property.structurallyDamaged = 0;
game[0].board[index].property.preDamagePurchasePrice = 0;
game[0].board[index].property.preDamageBaseRent = 0;
game[0].board[index].property.maintenanceCostMultiplierPercent = 100;
game[0].board[index].property.baseRent = 0;
game[0].board[index].property.houseCost = 0;
game[0].board[index].property.hotelCost = 0;
}

void setUtility(GameState game[], int index, char name[], int purchasePrice)
{
    game[0].board[index].index = index;
    game[0].board[index].type = UTILITY;

    strcpy(game[0].board[index].name, name);
    strcpy(game[0].board[index].property.name, name);

    game[0].board[index].property.purchasePrice = purchasePrice;
    game[0].board[index].property.mortgageValue = purchasePrice / 2;
    game[0].board[index].property.owner = -1;
    game[0].board[index].property.mortgaged = 0;
    game[0].board[index].property.loanLocked = 0;

    game[0].board[index].property.houses = 0;
    game[0].board[index].property.hotel = 0;
    game[0].board[index].property.insurance = NO_INSURANCE;
    game[0].board[index].property.insuranceRoundsLeft = 0;
    game[0].board[index].property.damaged = 0;
    game[0].board[index].property.repairCostOwed = 0;
    game[0].board[index].property.lostIncomeRoundsLeft = 0;
    game[0].board[index].property.age = 0;
    game[0].board[index].property.depreciation = 0;
    game[0].board[index].property.condition = 100;
    game[0].board[index].property.roundsSinceMaintenance = 0;
    game[0].board[index].property.structurallyDamaged = 0;
    game[0].board[index].property.preDamagePurchasePrice = 0;
    game[0].board[index].property.preDamageBaseRent = 0;
    game[0].board[index].property.maintenanceCostMultiplierPercent = 100;
    game[0].board[index].property.baseRent = 0;
    game[0].board[index].property.houseCost = 0;
    game[0].board[index].property.hotelCost = 0;
}

```

```

void setSpecialSquare(GameState game[],
                    int index,
                    SquareType type,
                    char name[])
{
    game[0].board[index].index = index;
    game[0].board[index].type = type;

    strcpy(game[0].board[index].name, name);

    /* Owner must be -1 (unowned), not the default 0, otherwise this
       square would look like it belongs to Player 0. */
    game[0].board[index].property.owner = -1;
    game[0].board[index].property.mortgaged = 0;
    game[0].board[index].property.loanLocked = 0;
    game[0].board[index].property.houses = 0;
    game[0].board[index].property.hotel = 0;
    game[0].board[index].property.insurance = NO_INSURANCE;
}

void initializeBoard(GameState game[])
{
    setSpecialSquare(game, 0, GO, "GO");
    setSpecialSquare(game, 2, COMMUNITY_FUND, "Community Development Fund");
    setSpecialSquare(game, 4, TAX, "Income Tax");
    setSpecialSquare(game, 7, EVENT, "National Event Card");
    setSpecialSquare(game, 10, JAIL, "Jail / Just Visiting");
    setSpecialSquare(game, 17, INSURANCE, "Sri Lanka Insurance");
    setSpecialSquare(game, 20, FREE_PARKING, "Free Parking");
    setSpecialSquare(game, 22, EVENT, "National Event Card");
    setSpecialSquare(game, 30, GO_TO_JAIL, "Go To Jail");
    setSpecialSquare(game, 33, INSURANCE, "Ceylinco Insurance");
    setSpecialSquare(game, 36, EVENT, "National Event Card");
    setSpecialSquare(game, 38, BANK, "Bank of Ceylon");

    setRailway(game, 5, "Colombo Fort Railway Station", 1500);
    setRailway(game, 15, "Kandy Railway Station", 1500);
    setRailway(game, 25, "Galle Railway Station", 1500);
    setRailway(game, 35, "Jaffna Railway Station", 1500);

    setUtility(game, 12, "Ceylon Electricity Board", 1500);
    setUtility(game, 28, "National Water Supply and Drainage Board", 1500);

    /* Brown */
    setProperty(game, 1, "Pettah", BROWN, 1500, 100, 750, 500, 2000);
    setProperty(game, 3, "Maradana", BROWN, 1800, 120, 900, 500, 2000);

    /* Light Blue */
    setProperty(game, 6, "Bambalapitiya", LIGHT_BLUE, 2500, 180, 1250, 750, 3000);
    setProperty(game, 8, "Wellawatte", LIGHT_BLUE, 2700, 200, 1350, 750, 3000);
    setProperty(game, 9, "Mount Lavinia", LIGHT_BLUE, 3000, 220, 1500, 750, 3000);

    /* Pink */
    setProperty(game, 11, "Nugegoda", PINK, 3500, 260, 1750, 1000, 4000);
    setProperty(game, 13, "Maharagama", PINK, 3800, 280, 1900, 1000, 4000);
    setProperty(game, 14, "Kottawa", PINK, 4000, 300, 2000, 1000, 4000);

    /* Orange */
    setProperty(game, 16, "Negombo", ORANGE, 4500, 350, 2250, 1250, 5000);
    setProperty(game, 18, "Katunayake", ORANGE, 4700, 370, 2350, 1250, 5000);
    setProperty(game, 19, "Ja-Ela", ORANGE, 5000, 400, 2500, 1250, 5000);
}

```

```
/* Red */
setProperty(game, 21, "Kandy City", RED, 5500, 450, 2750, 1500, 6000);
setProperty(game, 23, "Peradeniya", RED, 5800, 480, 2900, 1500, 6000);
setProperty(game, 24, "Katugastota", RED, 6000, 500, 3000, 1500, 6000);

/* Yellow */
setProperty(game, 26, "Galle Fort", YELLOW, 6500, 600, 3250, 2000, 8000);
setProperty(game, 27, "Unawatuna", YELLOW, 6800, 620, 3400, 2000, 8000);
setProperty(game, 29, "Hikkaduwa", YELLOW, 7000, 650, 3500, 2000, 8000);

/* Green */
setProperty(game, 31, "Jaffna Town", GREEN, 8000, 750, 4000, 2500, 10000);
setProperty(game, 32, "Nallur", GREEN, 8300, 780, 4150, 2500, 10000);
setProperty(game, 34, "Trincomalee", GREEN, 8500, 800, 4250, 2500, 10000);

/* Dark Blue */
setProperty(game, 37, "Nuwara Eliya", DARK_BLUE, 10000, 1000, 5000, 3000, 12000);
setProperty(game, 39, "Galle Face", DARK_BLUE, 12000, 1200, 6000, 3000, 12000);
}
```